Appendix D: Details of Lemmas 11-13

Let $g_1(\mathbf{z}) = \lambda \sum_{j=1}^{2d_\theta} \mathbf{z}_j^p$ and $g_2(\mathbf{z}) = \frac{1}{2}\mathbf{z}^T(\mathbf{W} - \beta\mathbf{I})\mathbf{z}$. Note that $g(\mathbf{z}) = g_1(\mathbf{z}) + g_2(\mathbf{z})$.

Lemma 11. *If $\mathbf{z} + \mathbf{d} \geq 0$, $\mathbf{z}_j \geq L$ for every index j, and $0 \leq s \leq v < 1$, then $\|\nabla g(\mathbf{z} + s\mathbf{d}) - \nabla g(\mathbf{z})\|^2 \leq 2s^2\{[\lambda p(1-p)]^2[L(1-v)]^{2p-4} + \|\mathbf{W} - \beta\mathbf{I}\|^2\}\|\mathbf{d}\|^2$.*

Proof. We consider $g_1(\mathbf{z})$ and $g_2(\mathbf{z})$ separately. By the definition of $g_1(\mathbf{z})$, we have $\|\nabla g_1(\mathbf{z} + s\mathbf{d}) - \nabla g_1(\mathbf{z})\| \leq \int_0^1 \|\nabla^2 g_1(\mathbf{z} + ts\mathbf{d})s\mathbf{d}\|dt = s\lambda p(1-p)\int_0^1 \sqrt{\sum_{j=1}^{2d_\theta}(\mathbf{z}_j + ts\mathbf{d}_j)^{2p-4}\mathbf{d}_j^2}dt$. Since $\mathbf{z}_j + ts\mathbf{d}_j \geq (1-s)L$, for all index j, we have $\int_0^1 \sqrt{\sum_{j=1}^{2d_\theta}(\mathbf{z}_j + ts\mathbf{d}_j)^{2p-4}\mathbf{d}_j^2}dt \leq ((1-s)L)^{p-2}\|\mathbf{d}\|$. In conjunction with $s \leq v$ and $0 < p < 1$, we get $\|\nabla g_1(\mathbf{z} + s\mathbf{d}) - \nabla g_1(\mathbf{z})\|^2 \leq s^2[\lambda p(1-p)]^2[L(1-v)]^{2p-4}\|\mathbf{d}\|^2$.

For $g_2(\mathbf{z})$, we have $\|\nabla g_2(\mathbf{z} + s\mathbf{d}) - \nabla g_2(\mathbf{z})\|^2 \leq s^2\|\mathbf{W} - \beta\mathbf{I}\|^2\|\mathbf{d}\|^2$. Thus, $\|\nabla g(\mathbf{z} + s\mathbf{d}) - \nabla g(\mathbf{z})\|^2 \leq 2\left(\|\nabla g_1(\mathbf{z}+s\mathbf{d}) - \nabla g_1(\mathbf{z})\|^2 + \|\nabla g_2(\mathbf{z}+s\mathbf{d}) - \nabla g_2(\mathbf{z})\|^2\right) \leq 2s^2\left\{[\lambda p(1-p)]^2[L(1-v)]^{2p-4} + \|\mathbf{W} - \beta\mathbf{I}\|^2\right\}\|\mathbf{d}\|^2$. □

Remark Lemma 11 is the consequence of that $g(\mathbf{z})$'s gradient is Lipschitz continuous in the region that $\mathbf{z}_j \geq L(1-v)$ for all index j.

Lemma 12. *Suppose that $p \in (0,1)$, $\mathbf{z} + \mathbf{d} \geq \mathbf{0}$, $\mathbf{z} > \mathbf{0}$, and $0 < s < 1$. Then $\mathbf{d}^T\left(\nabla g(\mathbf{z}) - \nabla g(\mathbf{z} + s\mathbf{d})\right) \leq \frac{-s}{1-s}\mathbf{d}^T\nabla^2 g(\mathbf{z})\mathbf{d}$.*

Proof. Since $\mathbf{z} + s\mathbf{d} = (1-s)\mathbf{z} + s(\mathbf{z}+\mathbf{d}) > \mathbf{0}$, all the derivatives below are well defined. We consider $g_1(\mathbf{z})$ and $g_2(\mathbf{z})$ separately. For $g_1(\mathbf{z})$, we have $\mathbf{d}^T\left(\nabla g_1(\mathbf{z}) - \nabla g_1(\mathbf{z} + s\mathbf{d})\right) = \lambda p\sum_j \mathbf{d}_j\left[\mathbf{z}_j^{p-1} - (\mathbf{z}_j + s\mathbf{d}_j)^{p-1}\right]$, and $\frac{-s}{1-s}\mathbf{d}^T\nabla^2 g_1(\mathbf{z})\mathbf{d} = \frac{s}{1-s}\lambda p(1-p)\sum_j \mathbf{z}_j^{p-2}\mathbf{d}_j^2$. Thus, it suffices to show that $\mathbf{d}_j\left[\mathbf{z}_j^{p-1} - (\mathbf{z}_j + s\mathbf{d}_j)^{p-1}\right] \leq \frac{s}{1-s}(1-p)\mathbf{z}_j^{p-2}\mathbf{d}_j^2,\ \forall j$.

If $\mathbf{d}_j \geq 0$, by the mean-value theorem, there exists $\mathbf{z}_j' \in [\mathbf{z}_j, \mathbf{z}_j + s\mathbf{d}_j]$ such that $\mathbf{z}_j^{p-1} - (\mathbf{z}_j + s\mathbf{d}_j)^{p-1} = (1-p)(\mathbf{z}_j')^{p-2}s\mathbf{d}_j$. Since $p - 2 < 0$ and $\mathbf{z}_j' \geq \mathbf{z}_j > 0$, $\mathbf{d}_j\left[\mathbf{z}_j^{p-1} - (\mathbf{z}_j + s\mathbf{d}_j)^{p-1}\right] = s(1-p)(\mathbf{z}_j')^{p-2}\mathbf{d}_j^2 \leq s(1-p)\mathbf{z}_j^{p-2}\mathbf{d}_j^2 \leq \frac{s}{1-s}(1-p)\mathbf{z}_j^{p-2}\mathbf{d}_j^2$.

If $\mathbf{d}_j < 0$, let $v = -\frac{\mathbf{d}_j}{\mathbf{z}_j}$. Since $\mathbf{z}_j + \mathbf{d}_j \geq 0$ and $\mathbf{z}_j > 0$, we have $0 < v \leq 1$. Multiplying both sides of the desired inequality by $\frac{\mathbf{z}_j^{1-p}}{-\mathbf{d}_j} > 0$ gives $(1-sv)^{p-1} - 1 \leq \frac{sv}{1-s}(1-p)$. Let $u = sv$. Since $0 < s < 1$ and $0 < v \leq 1$, we have $0 < u < 1$. Moreover, $u \leq s$ implies $\frac{u}{1-u} \leq \frac{u}{1-s}$. It therefore suffices to show that $(1-u)^{p-1} - 1 \leq \frac{u(1-p)}{1-u}$. Define $\omega(u) = (1-u)^p - (1-u) - u(1-p)$. Then $\omega(0) = 0$ and $\omega'(u) = p\left[1 - (1-u)^{p-1}\right] < 0,\ 0 < u < 1$, because $p - 1 < 0$. Hence, $\omega(u) \leq \omega(0) = 0$, which is equivalent to $(1-u)^{p-1} - 1 \leq \frac{u(1-p)}{1-u}$. Consequently, $(1-u)^{p-1} - 1 \leq \frac{u(1-p)}{1-u} \leq \frac{u(1-p)}{1-s}$. Thus, $\mathbf{d}_j\left[\mathbf{z}_j^{p-1} - (\mathbf{z}_j + s\mathbf{d}_j)^{p-1}\right] \leq \frac{s}{1-s}(1-p)\mathbf{z}_j^{p-2}\mathbf{d}_j^2$. Combining the two cases and summing over j yields $\mathbf{d}^T\left(\nabla g_1(\mathbf{z}) - \nabla g_1(\mathbf{z} + s\mathbf{d})\right) \leq \frac{-s}{1-s}\mathbf{d}^T\nabla^2 g_1(\mathbf{z})\mathbf{d}$.

For $g_2(\mathbf{z})$, we have $\nabla g_2(\mathbf{z}) = (\mathbf{W} - \beta\mathbf{I})\mathbf{z}$, $\nabla^2 g_2(\mathbf{z}) = \mathbf{W} - \beta\mathbf{I}$. Therefore, $\nabla g_2(\mathbf{z} + s\mathbf{d}) = \nabla g_2(\mathbf{z}) + s(\mathbf{W} - \beta\mathbf{I})\mathbf{d}$, and hence $\mathbf{d}^T\left(\nabla g_2(\mathbf{z}) - \nabla g_2(\mathbf{z} + s\mathbf{d})\right) = s\mathbf{d}^T(\beta\mathbf{I} - \mathbf{W})\mathbf{d} \leq \frac{s}{1-s}\mathbf{d}^T(\beta\mathbf{I} - \mathbf{W})\mathbf{d} = \frac{-s}{1-s}\mathbf{d}^T\nabla^2 g_2(\mathbf{z})\mathbf{d}$, where the inequality follows from $\beta\mathbf{I} - \mathbf{W} \succeq \mathbf{0}$ and $0 < s < 1$. Therefore,

$$\mathbf{d}^T\left(\nabla g(\mathbf{z}) - \nabla g(\mathbf{z} + s\mathbf{d})\right) = \mathbf{d}^T\left(\nabla g_1(\mathbf{z}) - \nabla g_1(\mathbf{z} + s\mathbf{d})\right) + \mathbf{d}^T\left(\nabla g_2(\mathbf{z}) - \nabla g_2(\mathbf{z} + s\mathbf{d})\right)$$

$$\leq \frac{-s}{1-s}\mathbf{d}^T\nabla^2 g_1(\mathbf{z})\mathbf{d} + \frac{-s}{1-s}\mathbf{d}^T\nabla^2 g_2(\mathbf{z})\mathbf{d}$$
$$= \frac{-s}{1-s}\mathbf{d}^T\nabla^2 g(\mathbf{z})\mathbf{d}.$$

□

Remark. Lemma 12 follows from the special structure of $g(\mathbf{z})$, which consists of a separable L_p regularization term with $p \in (0,1)$ and a concave quadratic term.

Lemma 13. *If* $\Delta\phi \geq \frac{\beta}{2}\|\mathbf{d}\|^2$, *then* $\Delta\phi - s(\Delta\phi - \frac{\beta}{2}\|\mathbf{d}\|^2 + \tau\|\mathbf{d}\|^2) + s^2(\mu\|\mathbf{d}\|^2) \leq (1-s\delta)\Delta\phi$, *where* $\tau > 0, \mu > 0, 0 < \delta < \min\{\frac{2\tau}{\beta}, 1\}$, *and* $0 < s \leq \min\{\frac{1}{\mu}(\tau - \frac{\beta\delta}{2}), 1\}$.

Proof.

$$\begin{aligned}
&\Delta\phi - s\left(\Delta\phi - \frac{\beta}{2}\|\mathbf{d}\|^2 + \tau\|\mathbf{d}\|^2\right) + s^2\mu\|\mathbf{d}\|^2 - (1-s\delta)\Delta\phi \\
&= -s\left(\Delta\phi - \frac{\beta}{2}\|\mathbf{d}\|^2 + \tau\|\mathbf{d}\|^2\right) + s^2\mu\|\mathbf{d}\|^2 + s\delta\Delta\phi \\
&= s\left[\left(s\mu + \frac{\beta}{2} - \tau\right)\|\mathbf{d}\|^2 - (1-s\delta)\Delta\phi\right] \\
&\leq s\left[\left(s\mu + \frac{\beta}{2} - \tau\right)\|\mathbf{d}\|^2 - \frac{(1-\delta)\beta}{2}\|\mathbf{d}\|^2\right] \\
&= s\|\mathbf{d}\|^2\left(s\mu - \tau + \frac{\beta\delta}{2}\right),
\end{aligned}$$

where the inequality follows from $\frac{\beta}{2}\|\mathbf{d}\|^2 \leq \Delta\phi$. Due to $0 < \delta < \frac{2\tau}{\beta}$ and $s \leq \frac{1}{\mu}(\tau - \frac{\beta\delta}{2})$, we have $s\mu - \tau + \frac{\beta\delta}{2} \leq 0$. Therefore, $\Delta\phi - s\left(\Delta\phi - \frac{\beta}{2}\|\mathbf{d}\|^2 + \tau\|\mathbf{d}\|^2\right) + s^2(\mu\|\mathbf{d}\|^2) \leq (1-s\delta)\Delta\phi$. □